Numerical attractors for rough differential equations

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Norms and spaces

- For $p \geq 1$, $C^{p-\mathrm{var}}([a,b],H)$: the Banach space with finite norm

$$\|u\|_{p,[a,b]} = \|u_a\| + \interleave u \interleave_{p,[a,b]} = \|u_a\| + \interleave u \interleave_{p,[a,b]} = \left(\sup_{\Pi(a,b)} \sum_{[s,t]\in\Pi} |u_{s,t}|^p \right)^{1/p} < \infty.$$

- For $\beta \in (0,1)$, $C^{\beta-\mathrm{Hol}}([a,b],H)$ the Banach space equipped with norm

$$\|u\|_{\infty,\beta,[a,b]} := \|u\|_{\infty,[a,b]} + \interleave u \interleave_{\beta,[a,b]} = \|u\|_{\infty,[a,b]} + \sup_{a\leq s<t\leq b} \frac{\|u(t)-u(s)\|}{(t-s)^\beta}.$$

- For $\alpha, \beta \in (0,1)$, $C^{\alpha,\beta}([a_+,b],H)$, $C^{\alpha,\beta}([a,b_-],H)$ the Banach spaces equipped with norms

$$\begin{aligned} \|u\|_{\infty,\alpha,\beta,[a_+,b]} &:= \|u\|_{\infty,[a,b]} + \sup_{a<s<t\leq b} |s-a|^\alpha \frac{\|u(t)-u(s)\|}{(t-s)^\beta} \\ \|u\|_{\infty,\alpha,\beta,[a,b_-]} &:= \|u\|_{\infty,[a,b]} + \sup_{a\leq s<t<b} |b-t|^\alpha \frac{\|u(t)-u(s)\|}{(t-s)^\beta}. \end{aligned}$$

- One can prove that $\interleave y \interleave_{\alpha,\beta,[a_+,b]} = \sup_{a<s<b} (s-a)^\alpha \interleave y \interleave_{\beta,[s,b]}$, thus $C^\beta([a,b]) \subset C^{\beta,\beta}([a_+,b]) \cap C^{\beta,\beta}([a,b_-])$ with $\interleave u \interleave_{\beta,\beta,[a_+,b]} \leq |b-a|^\beta \interleave u \interleave_{\beta,[a,b]}$.

Rough paths

In the simplest form for $\beta \in (\frac{1}{3}, \frac{1}{2})$, a couple $\mathbf{x} = (x, \mathbb{X})$, with $x \in C^{\beta}([a,b], H)$ and $\mathbb{X} \in C_2^{2\beta}([a,b]^2, H \otimes H)$ is called a *rough path* if they satisfy Chen's relation

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = x_{s,u} \otimes x_{u,t}, \qquad \forall a \leq s \leq u \leq t \leq b. \tag{1.1}$$

We write $\mathbb{X}_{s,t} = (x \otimes x)_{s,t} =: \int_s^t x_{s,u} \otimes dx_u$. In a general form, $(y \otimes x)_{s,t} =: \int_s^t y_{s,u} \otimes dx_u$ satisfies Chen's relation

$$(y \otimes x)_{s,t} - (y \otimes x)_{s,u} - (y \otimes x)_{u,t} = y_{s,u} \otimes x_{u,t}.$$

Denote by $\mathcal{C}^{\beta}([a,b]) \subset C^{\beta} \oplus C^{2\beta}$ the set of all rough paths in $[a,b]$, then $\mathcal{C}^{\beta}$ is a closed set but not a linear space, equipped with the rough path semi-norm

$$\|\mathbf{x}\|_{\beta,[a,b]} := \|x\|_{\beta,[a,b]} + \|\mathbb{X}\|_{2\beta,[a,b]^2}^{\frac{1}{2}} < \infty. \tag{1.2}$$

$\beta > \frac{1}{2}$: Young integrals

- Consider $y \in C^\beta([a,b], V \otimes H)$ and $x \in C^\beta([a,b], H)$, there exists a Young integral $\int_a^b y dx$ defined by

$$\int_a^b y dx = \lim_{|\Pi| \to 0} \sum_{[s,t] \in \Pi} y_s x_{s,t},$$

- Young-Loeve (Y-L) estimate

$$\Big\| \int_s^t y_u dx_u - y_s x_{s,t} \Big\| \leq K_\beta |t-s|^{2\beta} \left\vert\kern-0.25ex\left\vert\kern-0.25ex\left\vert y \right\vert\kern-0.25ex\right\vert\kern-0.25ex\right\vert_{\beta,[a,b]} \left\vert\kern-0.25ex\left\vert\kern-0.25ex\left\vert x \right\vert\kern-0.25ex\right\vert\kern-0.25ex\right\vert_{\beta,[a,b]} .$$

In fact, we can prove using fractional calculus that

$$\Big\| \int_s^t y_u dx_u - y_s x_{s,t} \Big\| \leq K_\beta \Big(\|y_{s,\cdot}\|_{\infty,[s,t]} + \left\vert\kern-0.25ex\left\vert\kern-0.25ex\left\vert y \right\vert\kern-0.25ex\right\vert\kern-0.25ex\right\vert_{\beta,\beta,[s_+,t]} \Big) \Big(\|x_{\cdot,t}\|_{\infty,[s,t]} + \left\vert\kern-0.25ex\left\vert\kern-0.25ex\left\vert x \right\vert\kern-0.25ex\right\vert\kern-0.25ex\right\vert_{\beta,\beta,[s,t_-]} \Big).$$

$\beta \in (\frac{1}{3}, \frac{1}{2})$: Rough integrals

- A path $y \in C^\beta([a,b], V \otimes H)$ is then called to be *controlled by* $x \in C^\beta([a,b], H)$ if there exists a tube (y', R^y) with $y' \in C^\beta([a,b], V \otimes H \otimes H)$, $R^y \in C^{2\beta}(\Delta^2([a,b]), V \otimes H)$ such that

$$y_{s,t} = y'_s x_{s,t} + R^y_{s,t}, \qquad \forall \min I \le s \le t \le \max I.$$

  y' is called Gubinelli derivative of y, which is uniquely defined as long as $x \in C^\beta \setminus C^{2\beta}$ is truly rough (works for fBm B^H, $H \in (\frac{1}{3}, \frac{1}{2}]$ FH[Chapter 6]).
- Denote $F_{s,t} := y_s \otimes x_{s,t} + y'_s \mathbb{X}_{s,t}$. Then

$$\delta F_{s,u,t} := F_{s,t} - F_{s,u} - F_{u,t} = -y'_{s,u}\mathbb{X}_{u,t} - R^y_{s,u} \otimes x_{u,t}$$

$$\Rightarrow \quad \|\delta F_{s,u,t}\| \le |t-s|^{3\beta}\Big(\|x\|_{\beta,[s,t]} \left\|R^y\right\|_{2\beta,\Delta^2[s,t]} + \left\|y'\right\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]} \Big)$$

  Thanks to the sewing lemma, the rough integral $\int_s^t y_u dx_u$ can be defined as

$$\int_s^t y_u d\mathbf{x}_u := \lim_{|\Pi| \to 0} \sum_{[u,v]\in \Pi} [y_u \otimes x_{u,v} + y'_u \mathbb{X}_{u,v}]$$

  Moreover, one gets Y-L typed estimate:

$$\left\| \int_s^t y_u d\mathbf{x}_u - y_s x_{s,t} - y'_s \mathbb{X}_{s,t} \right\| \le C_\beta |t-s|^{3\beta}\Big(\|x\|_{\beta,[s,t]} \left\|R^y\right\|_{2\beta,\Delta^2[s,t]} + \left\|y'\right\|_{\beta,[s,t]} \|\mathbb{X}\|_{2\beta,\Delta^2[s,t]} \Big).$$

Existence and uniqueness theorem: RDEs

- Finite dimension: RDE $dy_t = f(y_t)dt + g(y_t)dx_t$ is understood in the integral form

$$y_t = y_0 + \int_0^t f(y_s)ds + \int_0^t g(y_s)d\mathbf{x}_s. \tag{1.3}$$

- Riedel and Scheutzow (2017): f is one sided Lipschitz and linear growth in perpendicular direction. $g \in C_b^3$ as usual. Solution of the form (y, y') with $y, y' \in C^\beta$, $y' = g(y)$, which is solved by Doss-Sussmann technique.
- Infinite dimension: consider rough evolution equation

$$dy_t = [Ay_t + f(y_t)]dt + g(y_t)dx_t, \tag{1.4}$$

with the solution understood in the mild form

$$y_t = S(t)y_0 + \int_0^t S(t-s)f(y_s)ds + \int_0^t S(t-s)g(y_s)d\mathbf{x}_s. \tag{1.5}$$

Key task: define $\int_0^t S(t-s)g(y_s)dx_s$ for $S(\cdot) \in C^{\beta,\beta}$.
- Garrido-Atienza & Lu & Schmalfuss (2015): define rough integral using fractional calculus and solution definition by Hu & Nualart (2009).
- Hesse & Neamtu (2019): define rough integrals for controlled rough path $y \in C^{\beta,\beta}$ and $x \in C^\beta$ using a modified sewing lemma.

Finite dimension case: Assumptions

- $(\mathbf{H}_f)$ $f : \mathbb{R}^d \to \mathbb{R}^d$ is locally Lipschitz continuous and of one-sided linear growth
$$\exists C > 0 : \langle y, f(y) \rangle \leq C(1 + \|y\|^2), \quad \forall y \in \mathbb{R}^d; \tag{1.6}$$
in addition f is of linear growth in the perpendicular direction, i.e. there exists $C_f > 0$ such that
$$\Big\| f(y) - \frac{\langle f(y), y\rangle}{\|y\|^2} y \Big\| \leq C_f \Big(1 + \|y\| \Big), \quad \forall y \neq 0; \tag{1.7}$$
- $(\mathbf{H}_g^b)$ g belongs to $C_b^3(\mathbb{R}^d, (\mathbb{R}^m, \mathbb{R}^d))$ such that
$$C_g := \max \Big\{ \|g\|_\infty, \|Dg\|_\infty, \|D_g^2\|_\infty, \|D_g^3\|_\infty \Big\} < \infty; \quad \text{or} \tag{1.8}$$
$(\mathbf{H}_g^l)$ $g(y) = Cy$ for $C \in ((\mathbb{R}^m, \mathbb{R}^d), \mathbb{R}^d)$ such that
$$C_g := \|C\| < \infty; \tag{1.9}$$
- $(\mathbf{H}_X)$ for a given $\nu \in (\frac{1}{3}, \frac{1}{2})$, x belongs to the space $C^\nu(\mathbb{R}, \mathbb{R}^m)$ of all continuous paths which are of finite $\nu-$Hölder norm on any interval $[s, t]$. In particular, x is a realization of a stochastic process $X_t(\omega)$ with stationary increments, such that x can be lifted into a realized component $\mathbf{x} = (x, \mathbb{X})$ of a stochastic process $(x_\cdot(\omega), \mathbb{X}_{\cdot,\cdot}(\omega))$ with stationary increments, and the estimate
$$E\Big(\|x_{s,t}\|^p + \|\mathbb{X}_{s,t}\|^q \Big) \leq C_{T,\nu} |t-s|^{p\nu}, \forall s, t \in [0, T] \tag{1.10}$$
holds for any $[0, T]$, with $p\nu \geq 1$, $q = \frac{p}{2}$ and some constant $C_{T,\nu}$.

Doss-Sussmann transformation

- LHD (2020): the solution $\phi_\cdot(\mathbf{x}, \phi_a)$ of $d\phi_u = g(\phi_u)dx_u, \quad u \in [a,b], \phi_a \in \mathbb{R}^d$ is C^1 w.r.t. ϕ_s, and $\frac{\partial \phi}{\partial \phi_a}(\cdot, \mathbf{x}, \phi_a)$ is the solution of the linearized system

$$d\xi_u = Dg(\phi_u(\mathbf{x}, \phi_a))\xi_u dx_u, \quad u \in [a,b], \xi_a = Id, \tag{1.11}$$

- Cass-Litterer-Lyons (2013): Greedy times $\tau_0 = \min I, \quad \tau_{i+1} := \inf\Big\{t > \tau_i : \|\mathbf{x}\|_{p-\mathrm{var},[\tau_i,t]} = \gamma\Big\} \wedge \max I$. Assign $N(\gamma, \mathbf{x}, I) := \sup\{i \in \mathcal{N} : \tau_i \leq \max I\}$.
- The Doss-Sussmann transformation $y_t = \phi_t(\mathbf{x}, z_t)$ there is an one-to-one correspondence between a solution y_t of (1.5) on a certain interval $[0,\tau]$ and a solution z_t of the associate ordinary differential equation

$$\dot z_t = \Big[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t)\Big]^{-1} f(\phi_t(\mathbf{x}, z_t)), \quad t \in [0,\tau],\ z_0 = y_0. \tag{1.12}$$

To estimate the solution norm growth, assign

$$\gamma_t := y_t - z_t, \quad \text{and} \quad \psi_t := \Big[\frac{\partial \phi}{\partial z}(t, \mathbf{x}, z_t)\Big]^{-1} - Id, \quad \forall t \in [0,\tau],$$

where $\tau > 0$ is chosen such that $16 C_p C_g \|\mathbf{x}\|_{p-\mathrm{var},[0,\tau]} \leq \lambda$ for some $\lambda \in (0,1)$ small enough.

Doss-Sussmann transformation

Lemma

- $g \in C_b^3$: *Assume that* $\phi(\cdot, \mathbf{x}, \phi_a)$ *is the solution of* $d\phi = g(\phi)dx$. *Introduce the semi-norm*

$$\|\kappa, R^\kappa\|_{p-\mathrm{var},[s,t]} := \|\kappa\|_{p-\mathrm{var},[s,t]} + \|R^\kappa\|_{q-\mathrm{var},[s,t]^2}.$$

Then for any interval $[a,b]$ *such that* $16 C_p C_g \|\mathbf{x}\|_{p-\mathrm{var},[a,b]} \leq 1$, *the following estimates hold*

$$i) \quad \left\|\phi, R^\phi\right\|_{p-\mathrm{var},[a,b]} \leq 8 C_p C_g \|\mathbf{x}\|_{p-\mathrm{var},[a,b]}; \tag{1.13}$$

$$ii) \quad \left\|\frac{\partial \phi}{\partial \phi_a}(t,\mathbf{x},\phi_a) - Id\right\|, \left\|\left[\frac{\partial \phi}{\partial \phi_a}(t,\mathbf{x},\phi_a)\right]^{-1} - Id\right\| \leq 16 C_p C_g \|\mathbf{x}\|_{p-\mathrm{var},[a,b]}. \tag{1.14}$$

- *The linear case* $g(y) = Cy$:

$$i) \quad \left\|\phi, R^\phi\right\|_{p-\mathrm{var},[a,b]} \leq 8 C_g \|\mathbf{x}\|_{p-\mathrm{var},[a,b]} \|\phi_a\|; \tag{1.15}$$

$$ii) \quad \left\|\frac{\partial \phi}{\partial \phi_a}(t,\mathbf{x},\phi_a) - Id\right\|, \left\|\left[\frac{\partial \phi}{\partial \phi_a}(t,\mathbf{x},\phi_a)\right]^{-1} - Id\right\| \leq 16 C_g \|\mathbf{x}\|_{p-\mathrm{var},[a,b]}. \tag{1.16}$$

Existence and uniqueness theorem

($\mathbf{H}_f'$) f is a locally Lipschitz continuous function which satisfies (2.13) and (1.7).

Theorem (S. Riedel, M. Scheutzow, 2017)

- *Under the assumptions* ($\mathbf{H}_f'$), ($\mathbf{H}_g^b$), ($\mathbf{H}_X$), *there exists a unique solution of* (1.5) *on any interval* $[0,T]$. *In addition, for each* $\lambda \in (0,1)$ *small enough, there exist some generic constants* $C(\lambda), \delta(\lambda)$ *such that the solution satisfies the following estimates*

$$\|y\|_{\infty,[0,T]} \leq e^{\delta(\lambda)T}\Big(\|y_0\| + C(\lambda)T + \frac{\lambda}{2}N\Big(\frac{\lambda}{16C_pC_g},\mathbf{x},[0,T]\Big)\Big) =: R. \tag{1.17}$$

- *Under the assumptions* ($\mathbf{H}_f'$), ($\mathbf{H}_g^l$), ($\mathbf{H}_X$), *there exists a unique solution of* (1.5) *on any interval* $[0,T]$. *In addition, for each* $\lambda \in (0,1)$ *small enough, there exist some generic constants* $C(\lambda), \delta(\lambda)$ *such that the solution satisfies*

$$\|y\|_{\infty,[0,T]} \leq \exp\{\delta(\lambda)T + \lambda N(\frac{\lambda}{16C_pC_g},\mathbf{x},[0,T])\}\Big(\|y_0\| + \frac{C(\lambda)}{\delta(\lambda)}\Big) - \frac{C(\lambda)}{\delta(\lambda)} =: R. \tag{1.18}$$

Condition (1.6) is equivalent to the following condition

$$\exists \bar{C} > 0 : \langle y, f(y)\rangle \leq \bar{C}\|y\|(1+\|y\|), \quad \forall y \in \mathbb{R}^d; \tag{1.19}$$

Generation of RDS: rough cocycles

- Finite dimension - Bailleul & Riedel & Scheutzow (2018)
- Infinite dimensional case: Garrido-Atienza & Lu & Schmalfuss (2010,2016), Hesse & Neamtu (2020)...

$T_1^2(\mathbb{R}^m) = 1 \oplus \mathbb{R}^m \oplus (\mathbb{R}^m \otimes \mathbb{R}^m)$, is the set with the tensor product

$(1,g^1,g^2)\otimes(1,h^1,h^2) = (1,g^1+h^1,g^1\otimes h^1+g^2+h^2), \quad \forall\, \mathbf{g} = (1,g^1,g^2), \mathbf{h} = (1,h^1,h^2) \in T_1^2(\mathbb{R}^m)$.

Then it can be shown that $(T_1^2(\mathbb{R}^m),\otimes)$ is a topological group with unit element $\mathbf{1} = (1,0,0)$. For $\beta \in (\frac{1}{p},\nu)$, denote by $\mathcal{C}^{0,\beta}([a,b],T_1^2(\mathbb{R}^m))$ the closure of $\mathcal{C}^\infty([a,b],T_1^2(\mathbb{R}^m))$ in $\mathcal{C}^\beta([a,b],T_1^2(\mathbb{R}^m))$, and by $\mathcal{C}_0^{0,\beta}(\mathbb{R},T_1^2(\mathbb{R}^m))$ the space of all $x:\mathbb{R}\to\mathbb{R}^m$ such that $x|_I \in \mathcal{C}^{0,\beta}(I,T_1^2(\mathbb{R}^m))$ for each compact interval $I\subset\mathbb{R}$ containing 0. Then $\mathcal{C}_0^{0,\beta}(\mathbb{R},T_1^2(\mathbb{R}^m))$ is equipped with the compact open topology given by the $p-$variation norm, i.e the topology generated by the metric:

$$d_\beta(\mathbf{x}_1,\mathbf{x}_2) := \sum_{k\geq 1}\frac{1}{2^k}(\|\mathbf{x}_1-\mathbf{x}_2\|_{\beta,[-k,k]}\wedge 1).$$

Let us consider a stochastic process $\bar{\mathbf{X}}$ defined on a probability space $(\bar{\Omega},\bar{\mathcal{F}},\bar{\mathbb{P}})$ with realizations in $(\mathcal{C}_0^{0,\beta}(\mathbb{R},T_1^2(\mathbb{R}^m)),\mathcal{F})$. Assume further that $\bar{\mathbf{X}}$ has stationary increments.

Generation of RDS: rough cocyles

Assign

$$\Omega := \mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m))$$

and equip with the Borel $\sigma-$ algebra $\mathcal{F}$ and let $\mathbb{P}$ be the law of $\bar{\mathbf{X}}$. Denote by θ the *Wiener-type shift*

$$(\theta_t\omega)_\cdot = \omega_t^{-1} \otimes \omega_{t+\cdot}, \forall t \in \mathbb{R}, \omega \in \mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m)), \tag{2.1}$$

and define the so-called *diagonal process* $\mathbf{X} : \mathbb{R} \times \Omega \to T_1^2(\mathbb{R}^m), \mathbf{X}_t(\omega) = \omega_t$ for all $t \in \mathbb{R}, \omega \in \Omega$. Due to the stationarity of $\bar{\mathbf{X}}$, it can be proved that θ is invariant under $\mathbb{P}$, then forming a continuous (and thus measurable) dynamical system on $(\Omega, \mathcal{F}, \mathbb{P})$ [BRSch17 -Theorem 5]. Moreover, $\mathbf{X}$ forms a β*-Hölder rough path cocycle*, namely, $\mathbf{X}_\cdot(\omega) \in \mathcal{C}_0^{0,\beta}(\mathbb{R}, T_1^2(\mathbb{R}^m))$ for every $\omega \in \Omega$, which satisfies the *cocyle relation*:

$$\mathbf{X}_{t+s}(\omega) = \mathbf{X}_s(\omega) \otimes \mathbf{X}_t(\theta_s\omega), \forall \omega \in \Omega, t, s \in \mathbb{R},$$

in the sense that $\mathbf{X}_{s,s+t} = \mathbf{X}_t(\theta_s\omega)$ with the increment notation $\mathbf{X}_{s,s+t} := \mathbf{X}_s^{-1} \otimes \mathbf{X}_{s+t}$. It is important to note that the two-parameter flow property

$$\mathbf{X}_{s,u} \otimes \mathbf{X}_{u,t} = \mathbf{X}_{s,t}, \forall s, t \in \mathbb{R}$$

is equivalent to the fact that $\mathbf{X}_t(\omega) = (1, x_t(\omega), \mathbb{X}_{0,t}(\omega))$, where $x_\cdot(\omega) : \mathbb{R} \to \mathbb{R}^m$ and $\mathbb{X}_{\cdot,\cdot}(\omega) : I \times I \to \mathbb{R}^m \otimes \mathbb{R}^m$ are random funtions satisfying Chen's relation relation (1.1).

Ergodicity of θ: $X = B^H$

- Garrido-Atienza, B. Schmalfuss (2011): the Wiener shift $\eta_t x_\cdot = x_{t+\cdot} - x_t$ is ergodic on $\Omega' = C^{0,\beta}(\mathbb{R}, \mathbb{R}^m)$ w.r.t. $\mathbb{P}_H$.
- Fiz-Hairer (2014, Theorem 10.4): there exists a full measure subset $\Omega_1 \subset \Omega'$ such that $\omega = (1, x, \mathbb{X}) \in \Omega$ for any $x \in \Omega_1$. Moreover, by Friz-Victoir (2010, Theorem 15.42, 45), one can choose this full measure subset Ω_1 such that it satisfies the piecewise linear approximations (mollifier approximation).
- Then consider the natural lift $\mathcal{S}$ on smooth paths

$$\mathcal{S}(x)_{s,t} = (1, x_{s,t}, \int_s^t x_{s,r} dx_r), \tag{2.2}$$

which can be extended to Ω_1 such that $\mathcal{S} : \Omega_1 \to \Omega$. By Bailleul, Riedel, Scheutzow (2017), there exists a metric dynamical system $(\hat{\Omega}, \hat{\mathcal{F}}, \hat{\mathbb{P}}, \theta)$ such that $\hat{\Omega} \subset \Omega$ and $\hat{\mathbb{P}} = \mathbb{P}^H \circ \mathcal{S}^{-1}$. Furthermore,

$$\theta_t \mathcal{S}(x) = \lim_{n\to\infty} \hat{\theta} \mathcal{S}(x^{(n)}) = \lim_{n\to\infty} \mathcal{S}(\eta_t x^{(n)}) = \mathcal{S}(\eta_t x). \tag{2.3}$$

- Since η is ergodic, by Garrido-Atienza, B. Schmalfuss (2011, Lemma 3), θ is also ergodic.

From now on, we assume additionally that $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ is ergodic.

Generation of RDS from rough equations

In particular, due to the fact that $\|\mathbf{X}_{\cdot}(\theta_h\omega)\|_{p-\mathrm{var},[s,t]} = \|\mathbf{X}_{\cdot}(\omega)\|_{p-\mathrm{var},[s+h,t+h]}$, it follows from Birkhorff ergodic theorem that

$$\Gamma(\mathbf{x},p) := \limsup_{n\to\infty}\Big(\frac{1}{n}\sum_{k=1}^{n}\|\theta_{-k}\mathbf{x}\|^p_{p-\mathrm{var},[-1,1]}\Big)^{\frac{1}{p}} = \Big(E\,\|\mathbf{X}_{\cdot}(\cdot)\|^p_{p-\mathrm{var},[-1,1]}\Big)^{\frac{1}{p}} = \Gamma(p) \qquad (2.4)$$

for almost all realizations $\mathbf{x}_t$ of the form $\mathbf{X}_t(\omega)$.

Theorem

Given the measurable metric dynamical system $(\Omega,\mathcal{F},\mathbb{P},\theta)$ *and the p- rough cocycle* $\mathbf{X}:\mathbb{R}\times\Omega\to T_1^2(\mathbb{R}^m)$ *as above, the system* (2.5) *generates a continuous random dynamical system* φ *over* $(\Omega,\mathcal{F},\mathbb{P},\theta)$*, such that for any* $[0,T]$ *and all* $\omega\in\Omega$, $\varphi(t,\omega)y_0$ *is the unique solution (in the Gubinelli sense) of* (1.5),

$$dy_t = f(y_t)dt + g(y_t)d\mathbf{X}_t(\omega), t\geq 0, \qquad (2.5)$$

on [0,T], where $\mathbf{x}=(x,\mathbb{X})$ *is the projection of* $\mathbf{X}_{\cdot}(\omega)$ *on* $\mathbb{R}^m\oplus(\mathbb{R}^m\otimes\mathbb{R}^m)$.

Proof

Given each realization ω of the diagonal process $\mathbf{X}_t(\omega) = \omega_t = (1, \mathbf{x}_t(\omega)) = (1, x_t(\omega), \mathbb{X}_{0,t}(\omega))$, we can define the stochastic integral in the pathwise sense

$$\int_a^b y_u d\omega_u := \int_a^b y_u d\mathbf{x}_u = \lim_{|\Pi|\to 0} \sum_{\Pi} \Big(y_u \otimes x_{u,v}(\omega) + y'_u \mathbb{X}_{u,v}(\omega) \Big).$$

The expression of the Darboux sum in the right hand side can be rewritten as

$$y_u \otimes x_{u,v}(\omega) + y'_u \mathbb{X}_{u,v}(\omega) =: (y_u, y'_u) \bar{\otimes} \Big(1, x_{u,v}(\omega), \mathbb{X}_{u,v}(\omega)\Big), \tag{2.6}$$

where the operator $\bar{\otimes}$ in the right hand side of (2.6) is defined as the left hand side. Since ω is the realization of $\mathbf{X}$, it follows from Chen's relation (1.1) that

$$\Big(1, x_{s,u}(\omega), \mathbb{X}_{s,u}(\omega)\Big) \otimes \Big(1, x_{u,v}(\omega), \mathbb{X}_{u,v}(\omega)\Big) = \Big(1, x_{s,v}(\omega), \mathbb{X}_{s,v}(\omega)\Big)$$

hence the shift property (2.1) yields

$$\Big(1, x_{u,v}(\omega), \mathbb{X}_{u,v}(\omega)\Big) = \omega_u^{-1} \otimes \omega_v = (\theta_u \omega)_{v-u}, \quad \forall 0 \le s \le t. \tag{2.7}$$

We therefore can rewrite the definition of the above rough integral as

$$\int_a^b y_u d\omega_u := \lim_{|\Pi|\to 0} \sum_{\Pi} (y_u, y'_u) \bar{\otimes} (\theta_u \omega)_{v-u}. \tag{2.8}$$

Proof

Since $\theta_{u+r}\omega = \theta_u \circ \theta_r \omega$, it is easy to check that the rough integral in (2.8) satisfies the additivity and the shift properties, i.e.

$$\int_a^c y_u d\omega_u = \int_a^b y_u d\omega_u + \int_b^c y_u d\omega_u, \quad \forall a \leq b \leq c; \tag{2.9}$$

$$\int_{a+r}^{b+r} y_u d\omega_u = \int_a^b y_{u+r} d(\theta_r\omega)_u, \quad \forall a \leq b, r \in \mathbb{R}. \tag{2.10}$$

These two properties (2.9), (2.10) and the existence and uniqueness theorem 2 then suffice to prove the cocycle property (**??**) of the generated random dynamical system from stochastic rough differential equation (2.5).

$$\begin{aligned}
\varphi(t+s,\omega)y_0 =& y_0 + \int_0^t f(\varphi(u,\omega)y_0)du + \int_0^t g(\varphi(u,\omega)y_0)d\omega_u \\
&+ \int_t^{t+s} f(\varphi(u,\omega)y_0)du + \int_t^{t+s} g(\varphi(u,\omega)y_0)dx_u \\
=& \varphi(t,\omega)y_0 + \int_0^s f(\varphi(u+t,\omega)y_0)du + \int_0^s g(\varphi(u+t,\omega)y_0)d(\theta_t\omega)_u \\
=& \varphi(s,\theta_t\omega)\varphi(t,\omega)y_0.
\end{aligned}$$

Random attractors for RDE: finite dimension

We would like to investigate the RDE of the form

$$dy_t = f(y_t)dt + g(y_t)dx_t. \tag{2.11}$$

($\mathbf{H}_f'$) and the dissipativity condition

$$\exists D_1 \geq 0, D_2 > 0 : \quad \langle y, f(y)\rangle \leq \|y\|(D_1 - D_2\|y\|), \quad \forall y \in \mathbb{R}^d, \tag{2.12}$$

Condition (2.12) is equivalent to the following condition: there exist constants $d_1 \geq 0, d_2 > 0$ such that

$$\langle y, f(y)\rangle \leq d_1 - d_2\|y\|^2, \quad \forall y \in \mathbb{R}^d; \tag{2.13}$$

($\mathbf{H}_g$) g belongs to $C_b^3(\mathbb{R}^d, (\mathbb{R}^m, \mathbb{R}^d))$ such that

$$C_g := \max\Big\{\|g\|_\infty, \|Dg\|_\infty, \|D_g^2\|_\infty, \|D_g^3\|_\infty\Big\} < \infty; \tag{2.14}$$

($\mathbf{H}_X$) for a given $\nu \in (\frac{1}{3}, \frac{1}{2})$, x belongs to the space $C^\nu(\mathbb{R}, \mathbb{R}^m)$ of all continuous paths which is of finite $\nu-$Hölder norm on any interval $[s,t]$. In particular, x is a realization of a stationary stochastic process $X_t(\omega)$, such that x can be lifted into a realized component $\mathbf{x} = (x, \mathbb{X})$ of a stationary stochastic process $(x_\cdot(\omega), \mathbb{X}_{\cdot,\cdot}(\omega))$, such that the estimate

$$E\Big(\|x_{s,t}\|^p + \|\mathbb{X}_{s,t}\|^q\Big) \leq C_{T,\nu}|t-s|^{p\nu}, \forall s,t \in [0,T] \tag{2.15}$$

holds for any $[0,T]$, with $p\nu \geq 1, q = \frac{p}{2}$ and some constant $C_{T,\nu}$.

Random attractors for RDE: finite dimension

($\mathbf{H}_{\mathcal{A}}$) There exists a duration $r > 0$ and constants $D_3 > 0$ of the deterministic system such that, for any starting point $y_0 \notin \mathcal{A}$, there exists a point $\mu_0 = \mu_0(y_0) \in \mathcal{A}$ satisfying

$$\|\mu_r(y_0) - \mu_r(\mu_0)\| \leq e^{-D_3}\|y_0 - \mu_0\|. \tag{2.16}$$

Theorem (LHD-DCDS 2022)

Assume that system (2.5) satisfies the assumptions $(\boldsymbol{H}_f')$, $(\boldsymbol{H}_g)$, $(\boldsymbol{H}_X)$, then there exists a random pullback attractor $\mathcal{A}(\omega)$ such that $|\mathcal{A}(\cdot)| \in^{\rho}$ for any $\rho \geq 1$. If in addition $(\boldsymbol{H}_{\mathcal{A}})$ and f is global Lipschitz continuous then the random attractor is upper semi-continuous with respect to the noise intensity in the sense that $\mathcal{A}(\omega) \to \mathcal{A}$ (w.r.t. the Hausdorff semi-distance) as $C_g \to 0$, both in the almost sure and in $\mathcal{L}^\rho$ senses. Moreover, if f is strictly dissipative then $\mathcal{A}(\omega)$ is a singleton provided that C_g is sufficiently small.

Sketch of the proof: Doss-Sussmann technique to conjugate the RDE to a random differential equations. Under the assumptions ($\mathbf{H}_f'$), ($\mathbf{H}_g$), ($\mathbf{H}_{\mathcal{X}}$), ($\mathbf{H}_{\mathcal{A}}$) and $f \in Lip$, the random attractor is upper semi-continuous, i.e.

$$\lim_{C_g \to 0} d_H\Big(\mathcal{A}(\omega)|\mathcal{A}\Big)^\rho = 0 \quad \text{a.s.} \quad \text{and} \quad \lim_{C_g \to 0} \mathbb{E}\, d_H\Big(\mathcal{A}(\cdot)|\mathcal{A}\Big)^\rho = 0, \quad \forall \rho \geq 1. \tag{2.17}$$

Compactness arguments

Lemma

For any $\lambda > 0$ small enough, there exist constants $\delta_\lambda, C_\lambda > 0$ such that for any solution μ_t of the ODE lying in the global attractor $\mathcal{A}$, the following estimates hold

$$g \in C_b^3 : \|y_t\| \leq \|y_0\| e^{-\delta_\lambda t} + C_\lambda N\Big(\frac{\lambda}{16 C_p C_g}, \mathbf{x}, [0,t]\Big); \tag{2.18}$$

$$g(y) = Cy : \|y_t\| \leq \exp\Big\{\lambda N\Big(\frac{\lambda}{16 C_p C_g}, \mathbf{x}, [0,t]\Big)\Big\}\Big[\|y_0\| e^{-\delta_\lambda t} + C_\lambda N\Big(\frac{\lambda}{16 C_p C_g}, \mathbf{x}, [0,t]\Big)\Big].$$

- To estimate $\|z_t\|$, we rewrite (1.12) as

$$\dot{z}_t = (Id + \psi_t) f(z_t + \gamma_t). \tag{2.19}$$

First, we are going to prove that there exists constants $\bar{C}_\lambda, \delta_\lambda > 0$ such that

$$\frac{d}{2dt}\|z_t\|^2 \leq \bar{C}_\lambda - \delta_\lambda \|z_t\|^2. \tag{2.20}$$

Indeed, consider two cases.

Proof

Case 1: $z_t + \gamma_t \neq 0$. From assumption ($\mathbf{H}_f$) and condition (**??**), we can check that

$$\begin{aligned}\frac{d}{2dt}\|z_t\|^2 =& \Big\langle z_t, (Id+\psi_t)\Big[\frac{\langle z_t+\gamma_t, f(z_t+\gamma_t)\rangle}{\|z_t+\gamma_t\|^2}(z_t+\gamma_t) + \pi^{\perp}_{z_t+\gamma_t}(f(z_t+\gamma_t))\Big]\Big\rangle \\ =& \underbrace{\Big\langle z_t, (Id+\psi_t)\frac{(z_t+\gamma_t)}{\|z_t+\gamma_t\|}\Big\rangle}_{=:M_1}\underbrace{\Big\langle \frac{z_t+\gamma_t}{\|z_t+\gamma_t\|}, f(z_t+\gamma_t)\Big\rangle}_{=:M_2} \\ &+ \underbrace{\Big\langle z_t, (Id+\psi_t)\pi^{\perp}_{z_t+\gamma_t}(f(z_t+\gamma_t))\Big\rangle}_{=:M_3}. \end{aligned} \tag{2.21}$$

we can show that there exists a generic constant $\bar{C}_\lambda > 0$ such that

$$M_1 M_2 \leq \bar{C}_\lambda - \frac{D_2}{2}(1-\lambda)\|z_t\|^2.$$

$$\begin{aligned} M_3 =& \Big\langle z_t+\gamma_t, \pi^{\perp}_{z_t+h_t}(f(z_t+\gamma_t))\Big\rangle - \Big\langle \gamma_t, \pi^{\perp}_{z_t+h_t}(f(z_t+\gamma_t))\Big\rangle \\ &+ \Big\langle z_t, \psi_t \pi^{\perp}_{z_t+h_t}(f(z_t+\gamma_t))\Big\rangle \\ \leq& (\|\gamma_t\| + \|\psi_t\|\|z_t\|)C_f(1+\|z_t+\gamma_t\|) \\ \leq& \bar{C}_\lambda + 2C_f\lambda\|z_t\|^2, \end{aligned}$$

Proof

Case 2: $z_t + \gamma_t = 0$. Then the same arguments show that

$$\begin{aligned}
\frac{d}{2dt}\|z_t\|^2 &= \langle z_t+\gamma_t, f(z_t+\gamma_t)\rangle - \langle \gamma_t, f(z_t+\gamma_t)\rangle + \langle z_t, \psi_t f(z_t+\gamma_t)\rangle \\
&= \Big\langle z_t+\gamma_t, f(z_t+\gamma_t)\Big\rangle - \langle \gamma_t, f(0)\rangle + \langle z_t, \psi_t f(0)\rangle \\
&\leq D_1\|z_t+\gamma_t\| - D_2\|z_t+\gamma_t\|^2 + (\|\gamma_t\| + \|\psi_t\|\|z_t\|)\|f(0)\| \\
&\leq \bar{C}_\lambda + \Big[2C_f\lambda - \frac{D_2}{2}(1-\lambda)\Big]\|z_t\|^2,
\end{aligned}$$

Hence (2.20) holds for all $z_t \in \mathbb{R}^d$ where $t \in [0, \tau \wedge \tau_{local}]$, with a generic constant $\bar{C}_\lambda$ and a sufficiently small $\lambda < 1$. This proves (2.20) by choosing

$$\delta_\lambda := \frac{D_2}{2}(1-\lambda) - 2C_f\lambda > 0 \quad \text{for} \quad 0 < \lambda < \frac{D_2}{D_2 + 4C_f} < 1.$$

Hence by Gronwall lemma and Cauchy inequality, one obtains

$$\|z_\tau\| \leq \|z_0\| e^{-\delta_\lambda \tau} + \frac{\bar{C}_\lambda}{\delta_\lambda} \Rightarrow \|y_\tau\| \leq \|z_\tau\| + \|\eta\| \leq \|y_0\| e^{-\delta_\lambda \tau} + \frac{\bar{C}_\lambda}{\delta_\lambda} + \frac{\lambda}{2}.$$

By constructing a greedy sequence of stopping times and using induction, one can easily show that

$$\|y_{\tau_i}\| \leq \|y_0\| \exp\Big\{-\delta_\lambda \tau_i\Big\} + iC_\lambda, \quad \forall i \in \mathcal{N}.$$

Euler scheme

For any finite partition $\Pi := \{0 = t_0 < t_1 < t_2 < \ldots < t_{m-1} < t_m = T\}$ such that $|\Pi| = \sup_k |t_{k+1} - t_k|$, we consider the explicit Euler scheme of equation (1.5) to approximate the fixed solution $y(\cdot, 0, y_0)$, i.e.,

$$\begin{aligned} y^{\Pi}_0 &= y_0; \\ y^{\Pi}_{k+1} &= y^{\Pi}_k + f(y^{\Pi}_k)(t_{k+1} - t_k) + g(y^{\Pi}_k)x_{t_k,t_{k+1}} + Dg(y^{\Pi}_k)g(y^{\Pi}_k)\mathbb{X}_{t_k,t_{k+1}}, \; 0 \leq k \leq m-1. \end{aligned} \tag{3.1}$$

Since the solution $y(t, 0, y_0)$ is fixed on $[0, T]$, its supremum norm is bounded by a fixed number R following (2.18).

Theorem (LHD, P. Kloeden)

Assume that $y(t, 0, y_0)$ is a solution of the rough differential equation (1.5) on $[0, T]$, under assumption ($\boldsymbol{H}_f$) for f, assumption ($\boldsymbol{H}^b_g$) or ($\boldsymbol{H}^l_g$) for g, and assumption ($\boldsymbol{H}_X$) for x. Then there exists a generic constant

$$C = C(f, g, \|\mathbf{x}\|_{\nu,[0,T]}, T, \|y_0\|) > 0$$

for R defined in (2.18) such that for $|\Pi| < \delta$ small enough

$$\sup_{0\leq k\leq m} \|y(t_k, 0, y_0) - y_k\| \leq C|\Pi|^{3\nu-1}. \tag{3.2}$$

Proof

Introduce a cutoff functions f_R such that

- $f_R(y) = f(y)$ for all $y \in B(0, R+1)$ and $f_R(y) = f(0)$ for all $\|y\| \geq R+2$;
- f_R is globally Lipschitz continuous w.r.t. constant C_{f_R} and is bounded by a constant $\|f_R\|_\infty$ on $\mathbb{R}^d$.

Specifically, $f_R(y) := f(\zeta_R(y))$ for all $y \in \mathbb{R}^d$, where $\zeta_R \in C^2(\mathbb{R}^d, \mathbb{R}^d)$ is constructed with $\zeta_R(y) = y$ if $\|y\| \leq R+1$ and $\zeta_R(y) = 0$ if $\|y\| \geq R+2$, such that ζ_R is bounded by $R+1$ and $\|D\zeta_R\|_\infty, \|D^2\zeta_R\|_\infty < \infty$.
Consider the truncated rough differential equation

$$dy_t = f_R(y_t)dt + g(y_t)dx(t), \ t \in [0, T]. \tag{3.3}$$

Next, define for the finite partition $\Pi := \{0 = t_0 < t_1 < t_2 < \ldots < t_{m-1} < t_m = T\}$ the explicit Euler scheme for the truncated rough differential equation (3.3) as follows

$$\begin{aligned} y^*_0 &= y_0; \\ y^*_{k+1} &= y^*_k + f_R(y^*_k)(t_{k+1} - t_k) + g(y^*_k)x_{t_k,t_{k+1}} + Dg(y^*_k)g(y^*_k)\mathbb{X}_{t_k,t_{k+1}}, 0 \leq k \leq m-1. \end{aligned} \tag{3.4}$$

Proof

Lemma

Assume that $\|f\|_\infty := \sup_{y\in\mathbb{R}^d}\|f(y)\| < \infty$*. Then under the assumptions* ($\mathbf{H}_g^b$) *and* ($\mathbf{H}_X$) *there exists a generic constant* $C_1 = C_1(\|f\|_\infty, C_g, \|\mathbf{x}\|_{\nu,[0,T]}, T)$ *independent of the initial condition, such that any solution* $y_\cdot$ *of* (1.5) *satisfies*

$$\|y\|_{p-\mathrm{var},[a,b]} \leq C_1(b-a)^\nu \quad \text{and} \quad \|R^y\|_{p-\mathrm{var},[a,b]} \leq C_1(b-a)^{2\nu}, \quad \forall 0 \leq a \leq b \leq T. \tag{3.5}$$

If in addition f *is globally Lipschitz continuous w.r.t. a constant* C_f *then there exists a generic constant* $C_2 = C_2(\|f\|_\infty, C_f, C_g, \|\mathbf{x}\|_{\nu,[0,T]}, T)$ *independent of the initial conditions such that any two solutions* $y^i_\cdot(\mathbf{x}, y^i_0), i = 1,2,$ *of equation* (1.5) *satisfy*

$$\|y^2 - y^1\|_{\infty,[a,b]} \leq C_2\|y^2_a - y^1_a\|, \quad \forall 0 \leq a \leq b \leq T. \tag{3.6}$$

Proof

Namely, denote by z_k the solution to (3.3) at time T that starts from y_k^* at time t_k, i.e. $z_k := y_R(T,t_k,y_k^*)$. Then $z_0 = y_R(T,0,y_0^*) = y(T,0,y_0)$ and $z_m = y_R(T,t_m,y_m^*) = y_m^*$. Using (3.6), we obtain

$$
\begin{aligned}
\|y_R(T,0,y_0^*) - y_m^*\| &\leq \sum_{k=0}^{m-1} \|z_k - z_{k+1}\| \\
&\leq \sum_{k=0}^{m-1} \|y_R(T,t_k,y_k^*) - y_R(T,t_{k+1},y_{k+1}^*)\| \\
&\leq \sum_{k=0}^{m-1} \|y_R(T,t_{k+1},y_R(t_{k+1},t_k,y_k^*)) - y_R(T,t_{k+1},y_{k+1}^*)\| \\
&\leq C_2(\|f_R\|_\infty, C_{f_R}, C_g, \|\mathbf{x}\|_{\nu,[0,T]}, T) \sum_{k=0}^{m-1} \|y_R(t_{k+1},t_k,y_k^*) - y_{k+1}^*\|. \qquad (3.7)
\end{aligned}
$$

Proof

On the other hand, from the definition of $y_R(t_{k+1}, t_k, y_k^*)$ and y_{k+1}^*

$$
\begin{aligned}
&\|y_R(t_{k+1}, t_k, y_k^*) - y_{k+1}^*\| \\
\leq & \Big| \int_{t_k}^{t_{k+1}} [f_R(y_R(u)) - f_R(y_k^*)]du + \int_{t_k}^{t_{k+1}} [g(y_R(u)) - g(y_k^*)]dx_u \Big| \\
\leq & C_1(\|f_R\|_\infty, C_g, \|\mathbf{x}\|_{\nu,[0,T]}, T)\Big[C_{f_R} T^{1-2\nu} + C_p\Big(\|\mathbf{x}\|_{\nu,[0,T]} + \|\mathbf{x}\|^2_{\nu,[0,T]} \Big)\Big](t_{k+1} - t_k)^{3\nu}.
\end{aligned}
$$

Therefore, one can estimate (3.7) with a constant

$$
C_3 = C_2 C_1 \Big[C_{f_R} T^{1-2\nu} + C_p\Big(\|\mathbf{x}\|_{\nu,[0,T]} + \|\mathbf{x}\|^2_{\nu,[0,T]} \Big)\Big] \tag{3.8}
$$

as follows

$$
\|y_R(T, 0, y_0^*) - y_m^*\| \leq C_3 \sum_{k=0}^{m-1} (t_{k+1} - t_k)^{3\nu} \leq C_3 |\Pi|^{3\nu-1} \sum_{k=0}^{m-1} (t_{k+1} - t_k) = C_3 T |\Pi|^{3\nu-1}. \tag{3.9}
$$

The right hand side of (3.9) converges to zero as $|\Pi| \to 0$.

Proof

Hence one obtains (3.2) for the Euler numerical scheme of the truncated equation (3.3) by assigning

$$C(f,g,|||\mathbf{x}|||_{\nu,[0,T]},T,\|y_0\|) := C_3 T = C_2 C_1 \Big[C_{f_R} T^{1-2\nu} + C_p\Big(|||\mathbf{x}|||_{\nu,[0,T]} + |||\mathbf{x}|||^2_{\nu,[0,T]}\Big)\Big] T, \quad (3.10)$$

which depends on $f, g, \mathbf{x}$ and R, thus also on $\|y_0\|$ due to (2.18). Note that $y_R(t_k,0,y_0) = y(t_k,0,y_0)$, thus by choosing $|\Pi| < \delta$ for $\delta = \delta(\|f_R\|_\infty, C_{f_R}, C_g, |||\mathbf{x}|||_{\nu,[0,T]}, T)$ small enough, we deduce that

$$\sup_{0\leq k\leq m} \|y^*_k\| \leq \|y\|_{\infty,[0,T]} + C\delta^{3\nu-1} \leq R+1.$$

As a result, $f_R(y^*_k) = f(y^*_k)$ for $0 \leq k \leq m$, hence the Euler scheme (3.4) for the truncated equation (3.3) coincides with the actual Euler scheme (3.1) in the ball $B(0,R+1)$, which proves (3.2).
The conclusion still holds for the linear diffusion function g under assumption ($\mathbf{H}^l_g$), since one can introduce a bounded function g_R in a similar way to f_R, where R is given by (1.18). Since similar arguments are invovled, we skip the proof for this case here.

RDS from Euler scheme

we consider the explicit Euler scheme for the regular grid with step size $h > 0$, i.e. $\Pi = \{kh\}_{k \in \mathcal{N}}$ and

$$\begin{aligned} y_0^h &\in \mathbb{R}^d, \\ y_{k+1}^h &= y_k^h + f(y_k^h)h + g(y_k^h)x_{kh,(k+1)h}(\omega) + Dg(y_k^h)g(y_k^h)\mathbb{X}_{kh,(k+1)h}(\omega), \quad k \in \mathcal{N}. \end{aligned} \tag{3.11}$$

Such a scheme is well defined. Using (2.6) and (2.7), we rewrite (3.11) as

$$\begin{aligned} y_{k+1}^h &= \underbrace{\Big(y_k^h + f(y_k^h)h\Big)}_{=:F(h,y_k^h)} + \Big\langle \underbrace{\Big(g(y_k^h), Dg(y_k^h)g(y_k^h)\Big)}_{=:G(y_k^h)}, \Big(x_{kh,(k+1)h}(\omega), \mathbb{X}_{kh,(k+1)h}(\omega)\Big)\Big\rangle \\ &= F(h,y_k^h) + G(y_k^h)\bar{\otimes}\Big(1, x_{kh,(k+1)h}(\omega), \mathbb{X}_{kh,(k+1)h}(\omega)\Big) \\ &= F(h,y_k^h) + G(y_k^h)\bar{\otimes}\big(\theta_{kh}\omega\big)_h = H(h,\theta_{kh}\omega)y_k^h, \end{aligned} \tag{3.12}$$

where we introduce the generator function

$$H(h,\omega)y := F(y) + G(y)\bar{\otimes}\omega_h. \tag{3.13}$$

Discrete-time random dynamical system $\varphi^h : \mathcal{N}^h \times \Omega \times \mathbb{R}^d \to \mathbb{R}^d$ over $\mathcal{N}^h := \{kh\}_{k \in \mathcal{N}}$ and $(\Omega, \mathcal{F}, \mathbb{P}, \theta)$ such that for any $\omega \in \Omega$ and $y_0^h \in \mathbb{R}^d$, $\varphi^h(k,\omega)y_0^h$ is defined from (3.12) by

$$\begin{aligned} \varphi^h(0,\omega)y_0^h &\equiv y_0^h, \\ \varphi^h(kh,\omega)y_0^h &:= y_k^h = H(h,\theta_{(k-1)h}\omega) \circ \ldots \circ H(h,\omega)y_0^h, \quad \forall k \geq 1. \end{aligned} \tag{3.14}$$

Random attractor for Euler scheme

Theorem (LHD, P. Kloeden)

Under the hypotheses ($\mathbf{H}'_f$),($\mathbf{H}^b_g$) *and* ($\mathbf{H}_X$), *assume further that* f *is dissipative with* (2.12) *and of linear growth, i.e. there exists* C_f *such that*

$$\|f(y)\| \leq C_f(1 + \|y\|). \tag{3.15}$$

Then there exists a $h_0 > 0$ *such that for all* $h < h_0$, *the generated discrete-time random dynamical system* φ^h *from* (3.12) *admits a global random pullback attractor* $A^h(\omega)$.

Sketch of the proof. It suffices to prove that there exists an absorbing set for the generated RDS φ^h. Consider the Lyapunov function $\|y^h_k\|$ then by applying Cauchy inequality and using assumptions ($\mathbf{H}_f$), ($\mathbf{H}_g$) and (3.15) we obtain

$$\begin{aligned}\|y^h_{k+1}\|^2 =& \|y^h_k + f(y^h_k)h + g(y^h_k)x_{kh,(k+1)h} + Dg(y^h_k)g(y^h_k)\mathbb{X}_{kh,(k+1)h}\|^2 \\ \leq& \|y^h_k\|^2\Big(1 - D_2h + 2C_f^2h^2 + 4\chi h(1 + C_f^2h^2)\Big) + \xi^h_0\Big(\|\mathbf{x}(\omega)\|_{p-\mathrm{var},[kh,(k+1)h]}\Big)\end{aligned} \tag{3.16}$$

where

$$\xi^h_0(A) := \frac{D_1^2}{D_2}h + 2C_f^2h^2 + 2\chi C_f^2h^3 + 2(1 + \frac{1}{\chi h})\Big(C_g^2A^2 + C_g^4A^4\Big), \tag{3.17}$$

Proof

we show that for $h < h_0$

$$\|y^h_{k+1}\|^2 \leq \Big(1 - \frac{D_2}{4}h\Big)\|y^h_k\|^2 + \xi^h_0(\theta_{kh}\omega) < e^{-\frac{D_2}{4}h}\|y^h_k\|^2 + \xi^h_0(\theta_{kh}\omega).$$

Hence by induction we can prove that

$$\|y^h_k\|^2 \leq e^{-\frac{D_2}{4}hk}\|y^h_0\|^2 + \sum_{i=0}^{k-1} e^{-\frac{D_2}{4}ih}\xi^h_0(\theta_{k-i}\omega), \quad \forall k \geq 1.$$

We conclude that there exists a random pullback absorbing set $B^h(\omega) = B(0, R^h(\omega))$ where

$$R^h(\omega) := \sum_{k=0}^{\infty} e^{-\frac{D_2}{4}kh}\xi^h_0(\theta_{-kh}\omega). \tag{3.18}$$

Since $\xi^h_0 \in L^1$, so is $\log^+ \xi^h_0$, which implies that $\frac{1}{t}\log^+ \xi^h_0(\theta_{-t}\omega) \to 0$ as $t \to \infty$. Hence ξ^h_0 is tempered, and it follows that $R^h(\omega)$ is finite and also tempered a.s. This proves the existence of a random pullback attractor $A^h(\omega)$.

Upper semicontinuity of numerical attractor

Theorem (LHD,P. Kloeden)

Assume ($\boldsymbol{H}_g^b$), ($\mathbf{H}_X$) with the Gaussian process X and further that f is globally Lipschitz continuous and bounded, such that the dissipativity condition (2.12) is satisfied. Then φ^h admits a numerical pullback attractor $\mathcal{A}^h$ which converges to the attractor $\mathcal{A}$ a.s. in the Hausdorf semi-distance, i.e.,

$$\lim_{h\to 0} d_H(\mathcal{A}^h|\mathcal{A}) = 0 \quad a.s. \tag{3.19}$$

Sketch of the proof: For any time step $h < \frac{1}{2}$, assign $l := \lfloor \frac{1}{h} \rfloor \in \mathcal{N}$. Then $h \in (\frac{1}{l+1}, \frac{1}{l}]$ with $1 - h < lh \leq 1$. It implies from the proof of Theorem (6) that in case f is bounded, we obtain from (3.5) and (3.9) for $T := 1$ that there exist constants

$$\begin{aligned}
C_3(\omega) = C(\omega) =& C_2C_1\Big[C_f + C_p\Big(\|\mathbf{x}\|_{\nu,[0,1]} + \|\mathbf{x}\|^2_{\nu,[0,1]} \Big)\Big] \\
=& 5N\Big(\frac{1}{16C_pC_g}, \mathbf{x}, [0,1]\Big)^{\frac{2(p-1)}{p}} \Big[\|f\|_\infty + C_g \|\mathbf{x}\|_{\nu,[0,1]} \vee \Big(C_g \|\mathbf{x}\|_{\nu,[0,1]}\Big)^2\Big] \\
&\times \exp\Big\{4C_f + N\Big(\frac{1}{16C_pC_g(1+8\|f\|_\infty)}, \mathbf{x}, [0,1]\Big)\Big\} \\
&\times \Big[C_f + C_p\Big(\|\mathbf{x}\|_{\nu,[0,1]} + \|\mathbf{x}\|^2_{\nu,[0,1]} \Big)\Big]
\end{aligned} \tag{3.20}$$

Proof

such that

$$\begin{aligned}\|\varphi^h(lh,\omega)y_0-\varphi(1,\omega)y_0\| \leq & \|\varphi^h(lh,\omega)y_0-\varphi(lh,\omega)y_0\|+\|\varphi(lh,\omega)y_0-\varphi(1,\omega)y_0\| \\ \leq & C(\omega)h^{3\nu-1}+C_3(\omega)(1-lh)^\nu \\ \leq & C(\omega)h^{3\nu-1}.\end{aligned}$$

As a result, there exists a constant $C(\omega)$ independent of the initial condition y_0 such that

$$\|\varphi^h(lh,\omega)y_0\| \leq \|\varphi(1,\omega)y_0\|+C(\omega)h^{3\nu-1}, \quad \forall y_0\in\mathbb{R}^d. \tag{3.21}$$

Now applying Duc [**?**, Theorem 3.3] for dissipative function f, there exists a constant $\eta\in(0,1)$ and an integrable random variable $\xi_0(\omega)=\xi_0(\|\!|\mathbf{x}(\omega)|\!\|_{\nu,[0,1]})$ such that

$$\|\varphi(1,\omega)y_0\| \leq \eta\|y_0\|+\xi_0(\omega), \quad \forall y_0\in\mathbb{R}^d.$$

Hence

$$\|\varphi^h(lh,\omega)y_0\| \leq \eta\|y_0\|+\xi_0(\omega)+C(\omega)h^{3\nu-1},$$

which, by similar arguments to [**?**, Theorem 3.3] proves the existence of a pullback absorbing set $B^h(\omega)=B(0,R^h(\omega))$, where

$$R^h(\omega)=\sum_{k=0}^\infty \eta^k\Big(\xi_0(\theta_{-klh}\omega)+C(\theta_{-klh}\omega)h^{3\nu-1}\Big).$$

Proof

We can prove that

$$R^h(\omega) \quad \leq \quad 2\sum_{j=0}^{\infty}\eta^j\Big[\xi_0\Big(\left\vert\!\left\vert\!\left\vert\mathbf{x}(\theta_{-j}\omega)\right\vert\!\right\vert\!\right\vert_{\nu,[-1,1]}\Big)+C\Big(\left\vert\!\left\vert\!\left\vert\mathbf{x}(\theta_{-j}\omega)\right\vert\!\right\vert\!\right\vert_{\nu,[-1,1]}\Big)\Big] =: 2\bar{R}(\omega). \quad (3.22)$$

That means $A^h(\omega)\subset B^h(\omega)=B(0,R^h(\omega))$ for $h<\frac{1}{2}$ are entirely contained in a tempered set $B(0,2\bar{R}(\omega))$, hence uniformly attracted to $A(\omega)$ under φ. Hence $\forall\epsilon>0,\exists M(\epsilon,\omega)$ such that

$$d_H\Big(\varphi(k,\theta_{-k}\omega)A^h(\theta_{-k}\omega)|A(\omega)\Big)<\epsilon,\quad \forall k\geq M(\epsilon,\omega),\quad \forall h<\frac{1}{2}. \qquad (3.23)$$

With such fixed $M(\epsilon,\omega)$, there exists a constant $C(\omega,M)$ such that for all $h<\delta(\omega,M)\wedge\frac{1}{2}$ and all $y^h\in A^h$ in the ω-wise sense

$$\|\varphi^h(M,\theta_{-M}\omega)y^h(\theta_{-M}\omega)-\varphi(M,\theta_{-M}\omega)y^h(\theta_{-M}\omega)\|\leq C(\omega,M)h^{3\nu-1}<\epsilon.$$

Since the above inequality holds for all $y^h\in A^h$, it yields

$$d_H\Big(\varphi^h(M,\theta_{-M}\omega)A^h(\theta_{-M}\omega)|\varphi(M,\theta_{-M}\omega)A^h(\theta_{-M}\omega)\Big)\leq\epsilon. \qquad (3.24)$$

From (3.23) and (3.24), it follows from the invariance of A^h under φ^h and the triangular inequality

$$d_H(A^h(\omega)|A(\omega))<2\epsilon,\quad \forall h<\delta(\omega,M)\wedge\frac{1}{2}.$$

This proves that (3.19) hold almost surely.

Thank you!